

B.Tech Examination 2024-25
Department of Electronics and Communication Engineering
(Even Semester Regular & Supplementary)

Course Title: Principle Of Communication System, Course Code: ECEUGOE02

Full Marks: 80

Time: 3.00 Hrs

- Answer Question No. 1 and any 7 Questions from Question No. 2 to 12
- Figures on the right hand side margin indicate full marks.
- Answer all parts of a question in same place.
- Symbols have their usual meaning.

1. Answer any five questions.

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|---|---|
| (a) What is baseband transmission? | 2 |
| (b) What is modulation? | 2 |
| (c) Define transmission efficiency of an AM system. | 2 |
| (d) Define frequency deviation in FM. | 2 |
| (e) Write the properties of line coding. | 2 |
| (f) What is a lossless channel? | 2 |
| (g) State Shannon-Hartley Law of a communication channel. | 2 |

2. (a) Draw the block diagram of a communication system and briefly explain the function of each block. 5
- (b) Mention different types of modulation techniques. 2
- (c) Why do we need modulation in communication? 3

3. (a) What is meant by the term 'amplitude modulation'? 2
- (b) Define modulation index for AM wave. 2
- (c) Derive an expression for single-tone amplitude modulated wave. 4
- (d) Write the condition for over modulated AM wave and draw the waveform of an over modulated AM wave. 2

4. (a) Prove that the balanced modulator produces an output signal consisting of two side band only with the carrier removed. 5
- (b) Derive an expression of transmission efficiency of a single tone amplitude modulated signal. 3
- (c) Calculate the percentage of power savings for DSB-SC signal for the percentage modulation of 50%. 2

5. (a) What is frequency modulation? 2
- (b) Derive an expression for single tone frequency modulated signal. 4
- (c) Explain the generation of narrowband FM signals with suitable block diagram. 4

6. (a) What is Carson's rule? 2
- (b) Explain the operation of balanced slope detector. 5
- (c) A single tone frequency modulated wave is denoted by following expression
 $s(t) = 2 \cos[2\pi \times 10^6 t + 10 \sin(2\pi \times 10^3 t)]$. Determine carrier frequency, modulating frequency and maximum frequency deviation. 3

7. (a) State and Prove sampling theorem for a band limited signal. 6
- (b) What is aliasing effect? How is it removed? 2
- (c) Determine the Nyquist rate for a continuous time signal
 $x(t) = 6 \cos 50\pi t + 20 \sin 300\pi t - 10 \cos 100\pi t$ 2

8. (a) How is an analog signal converted to a digital signal in PCM system? 5
 (b) Derive the expression for transmission bandwidth in a PCM system. 3
 (c) A television signal having a bandwidth of 4.2 MHz is transmitted using binary PCM system. Given that number of quantization level is 512. Determine transmission bandwidth and final bit rate. 2
9. (a) Explain companding in non-uniform quantization? 4
 (b) Draw and explain compressor characteristics. 3
 (c) Describe about μ -law companding. 3
10. (a) Explain the operation of a delta modulator with suitable block diagram. 5
 (b) What is slope overload distortion and how is it avoided in ADM? 2
 (c) Given a sine wave of frequency f_m and amplitude A_m applied to a delta modulator having a step size Δ . Show that slope overload distortion will occur if $A_m > \frac{\Delta}{2\pi f_m T_s}$, Here T_s is the sampling frequency 3
11. (a) Consider the binary data sequence 1011101011. Draw the waveform for the following signalling format: (i) Unipolar RZ (ii) Split Phase Manchester coding (iii) AMI 3
 (b) Sketch the waveform of ASK, FSK and PSK signal for the input binary sequence 1101010010. 3
 (c) Explain briefly the coherent detection of BFSK signal with suitable block diagram. 4
12. (a) Define entropy, information rate and how are they related? 3
 (b) A discrete memory less source (DMS) X has five possible outcome with probabilities $P_1 = \frac{1}{2}$, $P_2 = \frac{1}{4}$, $P_3 = \frac{1}{8}$, $P_4 = \frac{1}{16}$, $P_5 = \frac{1}{16}$. Find entropy and information rate if there are 16 outcomes per second. 4
 (c) A DMS X has five symbols x_1, x_2, x_3, x_4 and x_5 with $P(x_1) = 0.4$, $P(x_2) = 0.19$, $P(x_3) = 0.16$, $P(x_4) = 0.15$ and $P(x_5) = 0.1$. Construct a Huffman Code for the source. 3

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